Generative Inverse Design of Solid-State Fluoride-Ion Electrolytes in Data-Sparse Chemical Space

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Abstract

Development of high-energy-density fluoride-ion batteries (FIBs) is critically limited by the scarcity of suitable solid-state electrolytes, as fluoride-ion conductors remain sparsely represented in existing materials databases compared with the well-explored lithium-ion chemical space. In this work, we employ a diffusion-based generative foundation model, to perform hypothesis-free inverse design of novel fluoride solid electrolytes under data-sparse conditions. Structure generation is conditioned exclusively on thermodynamic stability, quantified by the energy above the convex hull, thereby avoiding imposed structural biases and enabling extrapolation beyond established fluorite- and tysonite-type prototypes. To elucidate the role of training data distribution in generative performance, we compare candidates generated from a compositionally general dataset and a fluoride-restricted dataset. Despite identical generation and screening protocols, the hierarchical generation-validation workflow yields a compact set of viable fluoride solid-electrolyte candidates predominantly originating from the fluoride-focused dataset, highlighting the importance of domain-aligned training data in sparsely populated chemical spaces. Interfacial reaction-energy analysis further identifies $CsMoF_6$, $CsWF_6$, $KMoF_6$, KWF_6, MoF_6, and $RbWF_7$ as comparatively promising electrolytes, exhibiting near-neutral interfacial thermodynamics with representative electrode materials, while interfaces involving Ce anodes are consistently unfavorable across the candidate set. This work represents the first application of generative inverse design to solid-state FIB electrolytes and establishes a general, stability-guided computational framework for exploring underrepresented solid-state ion-conducting materials.

1 Introduction

The advancement of next-generation battery is fundamentally limited by the lack of materials that can simultaneously deliver high energy density, intrinsic safety, and long-term scalability.[1–3] Although lithium-ion batteries (LIBs) remain the dominant commercial technology, their reliance on organic liquid electrolytes introduces critical safety concerns, most notably thermal runaway associated with lithium dendrite formation.[4] In parallel, the growing dependence on geographically concentrated resources such as Li, Mn, and Co has intensified supply-chain vulnerabilities.[5] Solid-state electrolytes were proposed as a partial solution.[6] However, most ceramic electrolytes still rely on lithium as the charge carrier, leaving resource scarcity largely unaddressed.[7]

As an alternative, fluoride-ion batteries (FIBs) have recently attracted increasing attention.[8] By employing fluoride ions as charge carriers, FIBs enable high-voltage electrochemistry and multi-electron conversion reactions, leading to theoretical volumetric energy densities approaching ~5000 $Wh \cdot L^{-1}$.[9] Moreover, FIB architectures are inherently compatible with inorganic solid electrolytes,

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offering substantial safety advantages by eliminating flammable liquid components.[10] Despite these promises, FIB development remains at an early stage, primarily hindered by the limited availability of fast fluoride-ion conductors that are simultaneously electronically insulating.[11] To date, research efforts have focused largely on binary fluorides such as LaF_3 and $PbSnF_4$, as well as simple solid-solution strategies aimed at reducing operating temperatures.[12–14] However, conventional discovery approaches based on incremental doping and substitution restrict exploration to narrow compositional and structural spaces. Consequently, the identification of new fluoride-ion solid electrolytes demands discovery strategies that extend beyond empirically guided compositional design.

Designing inorganic solid-state materials requires navigating a high-dimensional and strongly nonlinear relationship among composition, crystal structure, and functional properties. Traditional materials discovery has largely relied on forward modeling approaches, including density functional theory (DFT), high-throughput screening, and evolutionary structure searches.[15] Although these methods are effective for evaluating known or hypothesized candidates, they inherently confine exploration to pre-defined chemical compositions and structural prototypes, thereby limiting access to sparsely sampled regions of chemical space.

In response to these limitations, inverse design strategies have recently emerged as a powerful alternative, in which target properties or stability constraints are specified a priori and candidate structures satisfying these criteria are directly generated.[16] Machine-learning-based generative models enable this paradigm by learning joint probability distributions over compositions, lattice parameters, and atomic coordinates, thereby permitting probabilistic sampling beyond the finite candidate pools accessible to conventional workflows. Generative materials modeling has evolved rapidly, advancing from early generative adversarial network (GAN) and variational autoencoder (VAE) frameworks to more recent diffusion models.[17] Notably, diffusion-based approaches have demonstrated superior capability in generating diverse and physically plausible crystal structures by iteratively denoising coupled distributions of atomic species and spatial coordinates.

To translate recent advances in generative inverse design into practical materials discovery, a model must satisfy two critical requirements: broad chemical generality and explicit control over thermodynamic stability. MatterGen, a foundational crystal generative model recently introduced by Microsoft, [18] represents the current state of the art in this domain. Trained via joint diffusion on large-scale inorganic DFT datasets derived from the Materials Project[19] and Alexandria[20,21] MatterGen is designed to span a wide chemical space rather than being confined to specific compositional families or structural prototypes. A key capability of MatterGen lies in its support for conditional generation, which enables the direct targeting of thermodynamic stability through the energy-above-convex-hull metric. By imposing stability constraints at the generation stage, this strategy substantially alleviates the computational cost associated with downstream DFT screening. The effectiveness of this approach was recently demonstrated by Parida *et al.* in the context of lithium-ion batteries (LIBs), where 91 stable Li-containing compounds were identified, including two previously unknown high-capacity cathode materials. Notably, this study highlighted the ability of an off-the-shelf generative model to explore chemical spaces without task-specific retraining.[22] It is important to note, however, that the solid-state LIB materials landscape corresponds to a relatively dense data regime, supported by extensive compositional coverage and a rich diversity of known structural prototypes in existing databases. In stark contrast, solid-state FIB electrolytes reside in a highly sparse chemical domain, with only a limited number of reported compositions and structural archetypes. As a result, prior discovery efforts have been largely restricted to tysonite and fluorite types.[9,23] FIB electrolytes therefore constitute a stringent and physically meaningful testbed for generative inverse design, probing the capacity of foundation models to extrapolate into sparsely populated chemical regions that lie well beyond the reach of conventional discovery strategies.

In this work, we examine whether generative inverse design can effectively navigate the sparsely sampled fluoride-containing chemical space to accelerate the discovery of solid electrolyte candidates for fluoride-ion batteries (FIBs). We employ MatterGen under a minimal constraint framework, imposing only an energy-above-convex-hull condition and deliberately avoiding additional structural or property-based biases to enable hypothesis-free exploration. To elucidate the role of training data distribution in generative performance, we compare structures generated from the broadly representative Alex-MP-20 dataset from the Materials Project[19] and Alexandria[20,21] datasets with those obtained from a fluoride-focused subset of the Materials Project.[19] This comparison allows us to assess whether broad chemical coverage facilitates exploration of sparse chemistries or whether domain-restricted training yields higher sampling fidelity.[24] Generated candidates are subsequently screened for thermodynamic stability and systematically analyzed with respect to structural diversity and relevance to known fluoride-ion conducting motifs. To the best of our knowledge, this study represents the first application of generative inverse design to solid-state fluoride-ion electrolytes. By explicitly targeting a chemical domain that remains severely underrepresented in existing databases, this work provides a critical benchmark of the ability of foundation models to extrapolate beyond empirically accessible materials spaces and identify next-generation FIB electrolyte candidates.

2 Methods

2.1 Stability-Conditioned Generative Modeling

MatterGen was employed to generate fluoride-containing candidate materials using a parallelized sampling workflow based on two independent source datasets: Materials Project[19] and Alexandria.[20,21] To systematically examine the influence of training data generality on generative behavior, we deliberately varied the compositional scope of the training corpora. For the MP-derived dataset, a fluoride-specific subset was constructed by retaining only compounds containing fluorine. In contrast, the full Alexandria dataset was used without compositional filtering, encompassing both fluorinated and non-fluorinated compounds to represent a compositionally general chemical space.

Candidate generation was conditioned exclusively on thermodynamic stability to prioritize physical realizability and synthetic accessibility. Specifically, structures were generated under an energy-above-convex-hull constraint of $E_{\mathrm{hull}} \leq 0.05$ eV·atom^{-1}, ensuring that proposed candidates possessed favorable thermodynamic formation energies.[25]

2.2 Hierarchical Post-Generation Screening

Following generation, a multi-stage filtering pipeline was applied to identify stable, electronically insulating, and practically tractable fluoride-ion solid electrolyte candidates. First, to ensure structural quality, we implemented the MatterSim SUN (Stable, Unique, Novel) approach.[18] MatterSim is a machine-learning force field (MLFF) trained on large-scale first-principles data and used here to efficiently evaluate relative energetic stability across a wide range of inorganic compositions.[26] Candidates were required to satisfy a stability threshold of $E_{\mathrm{hull}} \leq 0.1$ eV·atom^{-1}. The uniqueness and novelty filters were subsequently applied to eliminate structurally redundant duplicates and remove materials already present in reference databases. Second, electronic insulation was assessed using the band gap (Eg) as a screening criterion. The Atomistic Line Graph Neural Network (ALIGNN) model [27] was employed to rapidly estimate band gaps directly from atomic structures, bypassing computationally intensive first-principles calculations. To ensure negligible electronic leakage and predominantly ionic transport behavior, candidates were required to satisfy band gaps (E_{g}) larger than 3.5 eV benchmarked against the established fluoride-ion solid electrolyte examples.[28] Third, compositional filters were applied to exclude elements incompatible with practical solid electrolyte chemistry. Candidates containing (i) noble gases, (ii) radioactive elements, or (iii) mercury were removed due to chemical inertness, regulatory constraints, or toxicity concerns.[29] In addition, oxygen- and hydrogen-containing compounds were excluded to restrict the search space strictly to pure fluoride frameworks. Finally, charge neutrality was enforced as a fundamental chemical validity criterion. A candidate composition was considered valid if its total formal charge summed to zero under plausible oxidation state assignments. For transition-metal-containing compounds, which may exhibit variable valency, structures were retained provided that charge neutrality could be achieved under any chemically reasonable combination of oxidation states. This step effectively removes compositions that are unlikely to form stable bulk phases and are prone to decomposition or phase separation.

2.3 First-Principles Validation

Candidates passing the post-generation filtering pipeline were further validated using first-principles calculations. Density functional theory (DFT) simulations were performed using the Vienna Ab initio Simulation Package (VASP). [30] The projector-augmented wave (PAW) method was used, and the generalized gradient approximation (GGA) Perdew-Burke-Ernzerhof (PBE) [31] method was adopted for the exchange-correlation functional. The plane-wave cutoff value was set 520 eV. The lattice parameters and atomic positions were fully relaxed until the residual forces on each atom below 0.03 eV$\cdot$ Å^{-1}. Following the structural relaxations, stability analysis was performed. Thermodynamic stability was then evaluated using the energy above the convex hull, candidate with $E_{\mathrm{hull}} \leq 0.02$ eV$\cdot$atom^{-1} to assess their thermodynamic stability.[25] To avoid the formation of oxide or hydroxide phases that can deteriorate electrolyte performance, an additional screening criterion was applied to exclude candidates containing oxygen or hydrogen. These secondary phases also strengthen metal-oxygen bonding, reduce fluoride-ion mobility, thereby lowering ionic conductivity[11]. In addition, elastic properties obtained from DFT calculations were used as a mechanical screening criterion. The elastic moduli were obtained using the Voigt-Reuss-Hill (VRH) averaging scheme.[32] To screen mechanically reasonable candidates, only those with positive Voigt and Reuss bounds for both the bulk and shear moduli were retained. Furthermore, the Hill-averaged bulk modulus (B_{H}) was required to exceed 16 GPa, comparable to representative fluoride solid electrolytes. Space-group symmetries were determined using the spglib library as implemented in pymatgen.[33] To ensure consistency between the assigned symmetry and the relaxed lattice geometry, an adaptive symmetry-tolerance scheme was used. The symmetry tolerance (symprec) was set to 0.10 Å and iteratively reduced in increments of 0.01 Å, down to a minimum of 0.001 Å. For each structure, the final space group was assigned using the smallest symprec value at which the symmetry-consistent cell reproduced the relaxed lattice metrics. An angular tolerance of 5° was applied throughout the analysis.

2.4 Thermodynamic Interfacial Reactivity Analysis

The interfacial reaction energy tendency between electrolyte and electrode was evaluated using a thermodynamic reaction driving force based on a closed-system convex-hull analysis. For each electrode-electrolyte pair, a reference convex hull was constructed from Materials Project entries within the combined chemical system using pymatgen.[33] Pseudo-binary reactions of the form xA + (1-x)B were considered, and for each composition the reaction energy was evaluated with respect to the equilibrium phases determined from the reference phase diagram. The maximum reaction driving force over the scanned composition range was used to quantify the interfacial reactivity. All calculations were performed at 0 K under closed-system conditions.

3 Result

Figure 1 illustrates the hierarchical filtering pipeline together with the resulting generated crystal structures. Materials generation and all subsequent screening steps were performed independently for the Alex-MP-20 and MP-F datasets, enabling a direct comparison of their respective efficiencies and outcomes. An initial set of structures was first screened based on thermodynamic stability using the convex-hull criterion and electronic insulation requirements. The candidate structure list was then restricted to fluoride-only compositions to preserve the intended fluoride-ion conduction chemistry. Subsequent filters removed structures containing excluded elements (noble gases, radioactive elements, or Hg) and energetically unfavorable phases. Finally, DFT-based structural validation was performed to confirm mechanical and chemical stability, resulting in a small number of final fluoride electrolyte candidates.

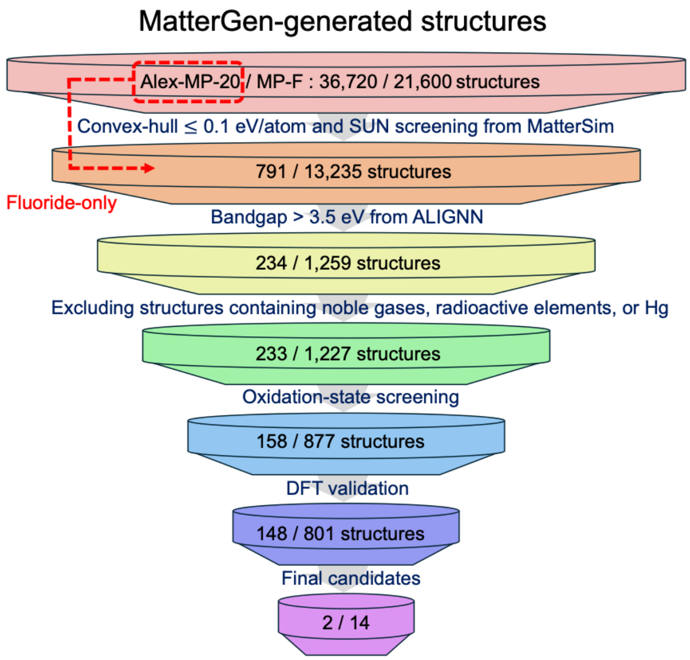

Figure 1. Crystal structure generation and hierarchical screening workflow, showing the progressive reduction from the initial generated structures to the final candidate materials for the Alex-MP-20 and MP-F datasets.

Figure 2 compares the elemental composition of fluoride-containing materials generated by MatterGen trained on two distinct datasets: the compositionally broad Alex-MP-20 dataset (**Figure 2a**) and the fluoride-restricted MP-F dataset (**Figure 2b**). The heatmaps display elemental counts only for candidates satisfying the MatterSim stability criterion ($E_{\mathrm{hull}} \leq 0.1$ eV·atom^{-1}).

The Alex-MP-20 trained model exhibits a broad elemental distribution across the periodic table (**Figure 2a**). Elements from alkali and alkaline-earth metals, transition metals, post-transition metals, and lanthanides all appear with significant frequency. This dispersion arises because the Alex-MP-20 dataset (a diverse collection of inorganic crystal structures with up to 20 atoms per primitive cell) was not curated with application-specific constraints. Consequently, the model maintains wide compositional coverage even after stability screening, avoiding bias toward specific chemical groups.

In contrast, the MP-F trained model produces a highly localized elemental distribution (**Figure 2b**). Counts are heavily concentrated within a small subset of elements prevalent in known fluoride compounds, while the majority of the periodic table shows negligible representation. This behavior reflects the inherent compositional bias of the fluoride-restricted training data, which reinforces established chemical motifs at the expense of exploration. These results illustrate a trade-off in generative behavior: while domain-specific training (MP-F) increases the sampling density of known fluoride-relevant chemistries, generalist training (Alex-MP-20) enables broader elemental exploration with minimal prior assumptions.

As shown in **Figures 2c and 2d**, the total number of generated structures differs between the two training datasets, with the Alex-MP-20 model yielding a larger candidate pool. Consequently, the higher absolute count of stable structures identified by MatterSim primarily reflects this upstream sampling volume rather than superior generative efficiency. The critical observation is that imposing electrolyte-relevant compositional constraints (specifically, restricting to fluorides) results in distinct candidate retention behaviors between the two training regimes, regardless of the initial generation size.

Because the Alex-MP-20 model is trained on a compositionally diverse dataset, a substantial fraction of its stability-screened candidates does not contain fluorine. Consequently, applying a fluoride-only compositional filter causes a sharp contraction in the candidate pool, as reflected by the reduced population in the band gap histogram (**Figure 2e**). In contrast, the MP-F model, trained on fluoride-restricted data, generates candidates that largely satisfy the fluoride criterion by design. As a result, its candidate pool is preserved more consistently during downstream screening (**Figure 2f**). notably, the normalized band gap distributions exhibit similar profiles in both cases. Therefore, the discrepancy in absolute counts observed in **Figure 2** arises primarily from candidate attrition due to the fluoride filter rather than qualitative shifts in electronic properties.

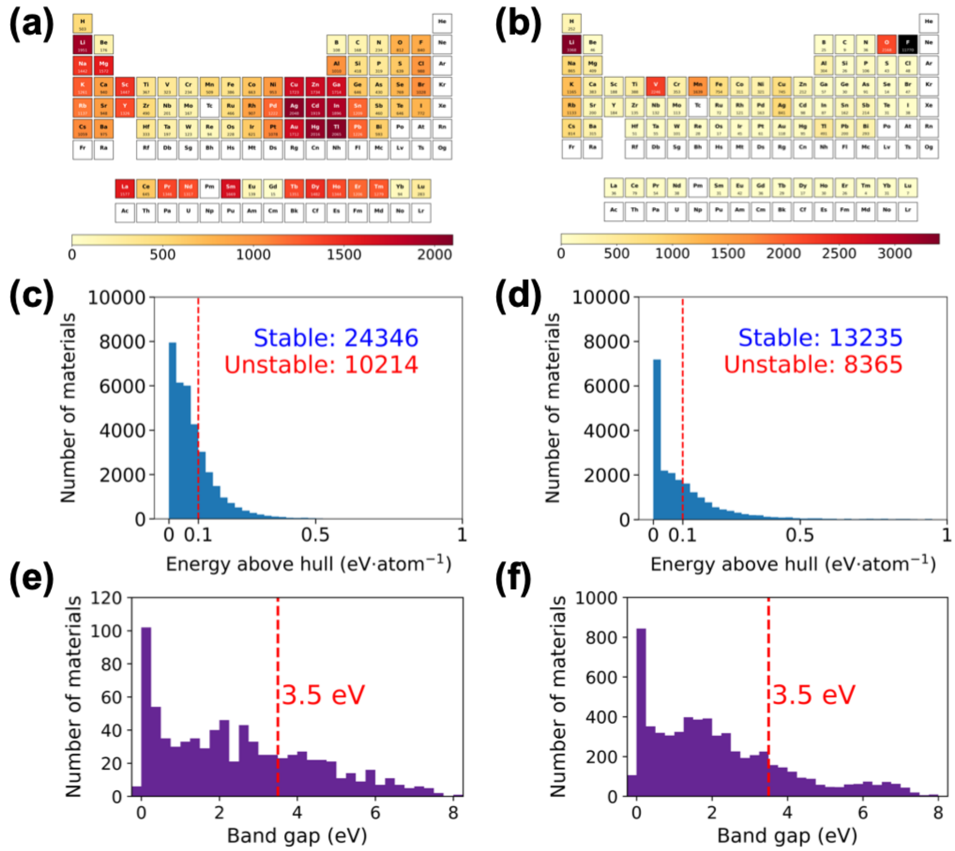

Figure 2. Periodic table heatmaps showing elemental counts in generated fluoride-containing materials for models trained on (a) the general Alex-MP-20 dataset and (b) the fluoride-restricted MP-F dataset. Histograms of MatterSim-predicted energy above hull (E_{hull}) for fluoride materials generated by models trained on (c) the Alex-MP-20 dataset and (d) the MP-F dataset. The dashed vertical line at 0.1 eV·atom^{-1} indicates the threshold for stability. Histograms of ALIGNN-predicted band gaps for stable fluoride candidates generated by models trained on (e) the Alex-MP-20 dataset and (f) the MP-F dataset. The red dashed line at 3.5 eV indicates the threshold for electronic insulation suitable for electrolyte applications.

Figure 3 presents parity plots comparing convex-hull energies (E_{hull}) predicted by MatterSim with DFT calculations. MatterSim exhibited a downward trend in E_{hull} relative to DFT, predicting lower energies across both datasets. Despite this deviation, the prediction accuracy is sufficient for first-stage screening, where the primary objective is to filter out highly unstable candidates prior to DFT validation. Quantitatively, the Alex-MP-20 dataset shows an RMSE of 0.06 eV·atom^{-1} and an MAE of 0.06 eV·atom^{-1}, while the MP-F dataset shows an RMSE of 0.07 eV·atom^{-1} and an MAE of 0.07 eV·atom^{-1}.

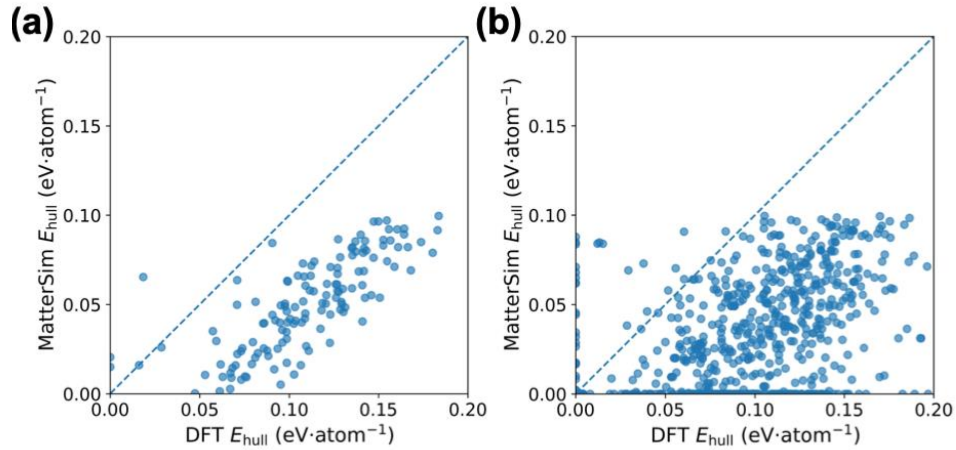

Figure 3. Parity plots comparing convex-hull energies predicted by MaterSim with DFT reference values for (a) the Alex-MP-20 candidate sets and (b) MP-F candidate sets. The dashed line represents perfect agreement (y = x)

Figure 4 presents the final candidate structures that satisfy the screening criteria for FIB solid-state electrolytes. All structures were fully relaxed using DFT calculations following the generative modeling and subsequent hierarchical filtering steps. Notably, only two candidates originated from the Alex-MP-20 dataset (**Figures 4a and 4b**), whereas the remaining structures (**Figures 4c-4p**) were derived from the MP-F dataset. Despite the use of identical generation and filtering protocols, the two datasets yielded markedly different numbers and types of viable structures, indicating a clear dependence of the generated candidate space on the compositional coverage and chemical diversity inherent to the initial training datasets.

To further assess the practical viability of the generated electrolyte candidates, their thermodynamic interfacial compatibility with representative electrode materials was evaluated. Electrolyte-electrode combinations exhibiting reaction energies close to zero are generally regarded as promising for subsequent detailed investigation. **Figure 5** presents a heatmap of closed-system interfacial reaction energies between the inverse-designed electrolyte candidates and representative cathode (BiF_3, FeF_3, and CuF_2) and anode (Sn, Pb, and Ce) materials.[8,34] Following established conventions, reaction energies near zero indicate thermodynamically stable interfaces, whereas increasingly negative values reflect a stronger driving force for interfacial decomposition.

Distinct electrode-dependent trends are observed. Among the cathode interfaces, BiF_3 maintains reaction energies closer to zero for multiple electrolyte candidates (predominantly blue regions), indicating comparatively favorable thermodynamic interfacial compatibility within the present materials set. In contrast, FeF_3 exhibits more negative reaction energies for most electrolyte combinations, suggesting a pronounced tendency toward interfacial decomposition. For the anode interfaces, Sn and Pb display intermediate reaction energies with limited variation across the electrolyte set, implying a relatively minor contribution to the overall interfacial reactivity trends. By comparison, Ce yields negative reaction energies for nearly all electrolytes (widespread red regions), indicating that interfacial mixing and decomposition are thermodynamically favored at Ce interfaces, despite the relatively low bulk instability of the electrolyte candidates themselves.

Considering these electrode-dependent trends, $CsMoF_6$, $CsWF_6$, $KMoF_6$, KWF_6, MoF_6, and $RbWF_7$ were identified as comparatively promising electrolyte candidates. These materials exhibited reaction energies closer to zero against BiF_3 and Sn (dark blue regions), corresponding to reduced thermodynamic driving forces for interfacial reactions and enhanced interfacial compatibility.

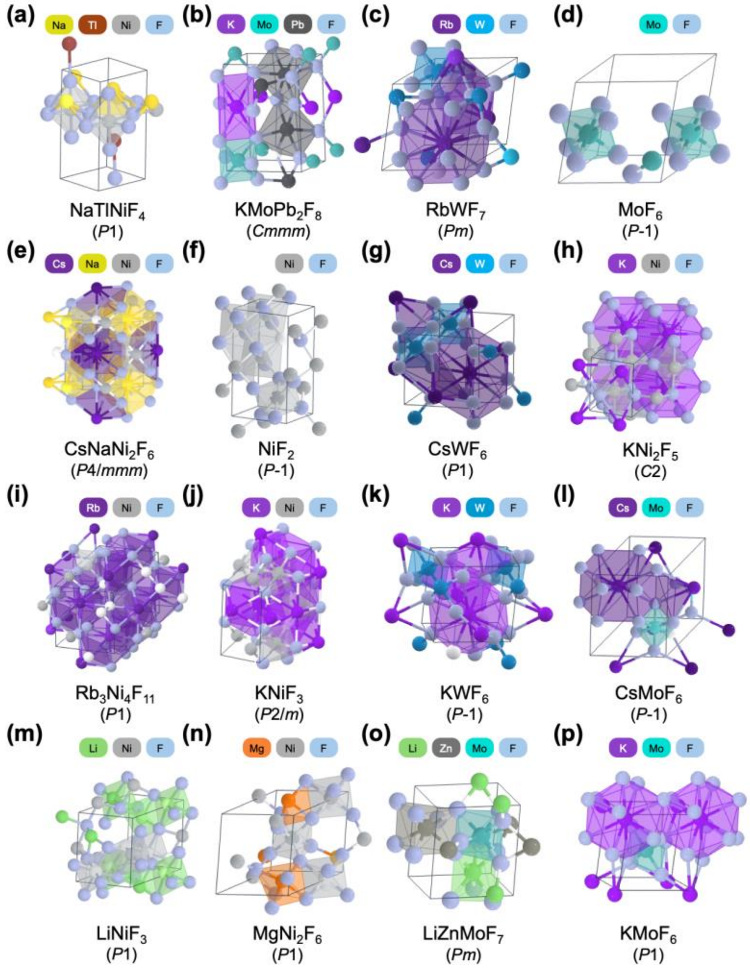

Figure 4. Optimized bulk crystal structures of FIB solid-state electrolytes obtained after the generation and hierarchical screening steps. Structures in panels (a,b) were predicted from the Alex-MP-20 dataset, whereas those in panels (c-p) were derived from the MP-F dataset.

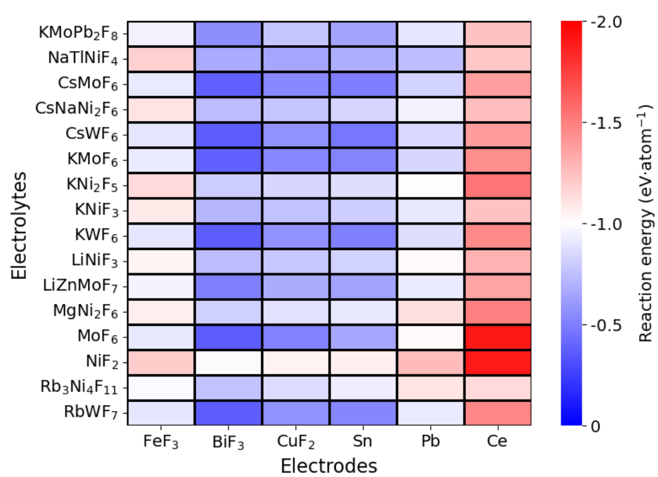

Figure 5. Heatmap of closed-system interfacial reaction energies for selected electrolyte-electrode combinations. Reaction energies close to zero (blue color) indicate thermodynamically stable interfaces, whereas increasingly negative values (red color) signify a stronger driving force for interfacial decomposition.

4 Discussion and Conclusions

In this work, we systematically examined the applicability of generative inverse design to solid-state fluoride-ion battery (FIB) electrolytes, a materials class that resides in a highly data-sparse chemical space. By employing MatterGen, a diffusion-based crystal generative model, under a minimal constraint framework conditioned solely on thermodynamic stability, we enabled hypothesis-free exploration beyond conventional fluorite- and tysonite-type electrolyte prototypes. This strategy allowed the generative process to sample structurally and compositionally diverse fluoride-containing candidates without imposing a priori structural biases.

A central outcome of this study is the clear demonstration that the characteristics of generative outputs are strongly governed by the underlying training data distribution. The comparison between candidates generated from a compositionally general dataset (Alex-MP-20) and a fluoride-restricted dataset (MP-F) reveals a fundamental trade-off between compositional generality and domain specificity. While the generalist model explored a broader elemental space, a substantial fraction of its candidates was eliminated during fluoride-specific post-generation filtering. In contrast, the fluoride-focused model produced a more compact and chemically aligned candidate pool, ultimately yielding a larger number of viable solid electrolyte structures after hierarchical screening. These results underscore that, in data-sparse domains, domain-aligned training data can significantly enhance sampling efficiency and downstream candidate retention, even when the same generative architecture and screening protocols are employed.

Through a multi-stage screening pipeline integrating MatterSim-based stability evaluation, machine-learning-assisted band gap prediction, compositional and charge-neutrality filtering, and first-principles DFT validation, we identified a limited set of fluoride-only solid electrolyte candidates that satisfy fundamental thermodynamic, electronic, and mechanical criteria. Importantly, the retained candidates span multiple space groups and exhibit diverse crystal structures, demonstrating that meaningful structural diversity can emerge even under strictly fluoride-only compositional constraints. This observation suggests that generative models, when coupled with physically motivated screening, are capable of uncovering nontrivial structural motifs in sparsely populated chemical spaces.

To further contextualize the practical relevance of the generated electrolytes, we performed a convex-hull-based interfacial reactivity analysis with representative electrode materials commonly considered in the FIB literature. Although this analysis is limited to closed-system thermodynamic considerations at 0 K, it provides qualitative insight into electrolyte-electrode compatibility trends. In particular, interfaces involving BiF_3 and Sn generally exhibit reduced thermodynamic driving forces for interfacial reactions, whereas Ce-based anodes consistently show unfavorable interfacial energetics across the candidate set. Based on these electrode-dependent trends, $CsMoF_6$, $CsWF_6$, $KMoF_6$, KWF_6, MoF_6, and $RbWF_7$ emerge as comparatively promising electrolyte candidates, exhibiting near-neutral interfacial reaction energies against selected electrodes.

Taken together, this study provides a systematic benchmark of stability-conditioned generative modeling applied to a sparsely populated fluoride-containing chemical domain. The results demonstrate that, when combined with hierarchical post-generation screening and first-principles validation, generative inverse design can yield chemically reasonable and physically plausible candidate materials even in regions of chemical space that are severely underrepresented in existing databases. Rather than replacing established first-principles approaches, generative modeling is shown here to function as an effective exploratory front end, narrowing the search space and guiding subsequent targeted investigations.

Looking forward, the present framework can be naturally extended beyond static stability analysis. In particular, molecular dynamics simulations based on machine-learning force fields offer a promising route to evaluate fluoride-ion transport properties over extended time and length scales. Such extensions will enable direct assessment of ionic conductivity and activation barriers, thereby bridging the gap between thermodynamic feasibility and functional performance. More broadly, the methodology established in this work provides a generalizable computational pathway for accelerating materials discovery in other underexplored solid-state ion-conducting systems.

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Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes] , [No] , or [N/A] .
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for NA).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While "[Yes] " is generally preferable to "[No] ", it is perfectly acceptable to answer "[No] " provided a proper justification is given (e.g., "error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering "[No] " or "[N/A] " is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "AI Co-Scientist Challenge Korea paper checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: Yes

   Justification: The main claims stated in the Abstract and Introduction accurately describe the application of stability-conditioned generative inverse design to solid-state FIB electrolytes and the analysis of training data distribution effects. These claims are directly supported by the results presented in the Results and interpreted in the Discussion and Conclusions

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: Yes

   Justification: The limitations of the study are explicitly discussed in the Discussion and Conclusions, including the reliance on thermodynamic stability as the primary screening criterion, the absence of explicit fluoride-ion transport calculations, and the use of closed-system interfacial thermodynamics at 0 K. The same section also outlines how these limitations may affect the scope of the conclusions and identifies future extensions to address them.

   Guidelines:

   - The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
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   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: N/A

   Justification: The paper does not present new theoretical theorems, analytical derivations, or formal proofs. All results are computational and data-driven, relying on established models and first-principles methods described in the Methods section and prior literature.

   Guidelines:

   - The answer NA means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented

by formal proofs provided in appendix or supplemental material.

- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: Yes

   Justification: The Methods section specifies the datasets used, generative model configuration, screening criteria, and all DFT and interfacial reaction-energy calculation parameters required to reproduce the main results. Together with the Results, these details provide a clear and sufficient pathway for independent reproduction using publicly available datasets and software.

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   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While AI Co-Scientist Challenge Korea does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
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     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: No

   Justification: The paper does not provide open access to code or scripts, as it relies on existing generative models and first-principles software subject to licensing and usage constraints. Nevertheless, detailed descriptions of datasets, model settings, and computational procedures are provided in the **Methods** section to support independent reproduction.

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- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
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- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: Yes

   Justification: The Methods section describes the datasets used, generative model settings, stability thresholds, and hierarchical screening criteria, as well as all DFT computational parameters required to interpret the results. These details are sufficient to understand the experimental setup and the basis of the reported conclusions, even though model training itself was not performed in this work.

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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